A Static-Universe Two-Channel Photon Propagation Model Confronted with DES-SN5YR

Distance Degeneracy, Universal Time Remapping, Chromatic Transfer, and Explicit Falsification Gates

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Abstract

We test whether the principal Type Ia supernova observables normally assigned to metric expansion can be reproduced by a non-expanding Euclidean geometry with two distinct photon-propagation channels. The forward channel is modeled as an inelastic, direction-preserving interaction that excites a transient local "wake" in the vacuum response. At effective-field level, the electromagnetic envelope obeys a wave equation with refractive perturbation χ, and the same local kernel $W \equiv \partial_t \ln n$ controls both frequency redshift and the separation of mutually incoherent photons. The model therefore gives $\Delta t_{\mathrm{obs}} = (1+z)\Delta t_{\mathrm{em}}$, or $b = 1$, without treating a supernova light curve as one coherent optical pulse. This is consistent with the DES-SN5YR measurement $b = 1.003 \pm 0.005\,(\mathrm{stat}) \pm 0.010\,(\mathrm{sys})$, while the deeper microscopic derivation of the wake response remains open.

A separate non-forward channel has $\sigma_{\mathrm{nf}} \propto E^{1/2} \propto \lambda^{-1/2}$ and produces wavelength-dependent beam depletion. Using the physically correct optical-depth convention, the DES Hubble-diagram fit corresponds to $\tau_{\infty,B} = 0.594$ at rest B. On the released DES-SN5YR distance vector, the resulting one-parameter static law remains nearly degenerate with one-parameter flat ΛCDM: $\chi^2 \simeq 1641.1$ versus 1640.1 on the default 1,768-object diagnostic sample, $\Delta\chi^2 = +1.1$, with both $\chi^2/\mathrm{dof} \simeq 0.93$. A full Hsiao-SED × measured-DECam-filter transfer calculation then falsifies an attractive alternative explanation: with the time remap disabled, chromatic erosion and passband slicing change fitted stretch by at most 5×10^{-4} and peak epochs by at most 0.05 d for $z \leq 0.8$. Thus amplitude erosion cannot fake the observed time dilation; it remains an independent chromatic discriminator.

We also report a preregistered species-clock test using the iron-group secondary maximum in raw DES i/z photometry. The result, $b_2 = 0.52 \pm 0.78$ from 120 detections, is inconclusive and is reported as such: it is consistent with both the universal remap and no dilation, because DES cadence, intrinsic bump diversity, and the $z \lesssim 0.3$ visibility window make the test underpowered. Finally, the model makes a sharply different surface-brightness prediction. Straight Euclidean rays give $D_A = D = (c/K)\ln(1+z)$ and a baseline $D_{L,0}/D_A = 1+z$, implying $I_{\mathrm{obs}} \propto (1+z)^{-2} e^{-\tau_\lambda}$. A preregistered static-frame analysis of the Lubin–Sandage Tolman data shows that their often-quoted I-band exclusion is strongly distance-frame dependent, but a 3.7σ $R-I$ exponent spread is approximately frame invariant and is presently the strongest empirical threat to the model. Draft 6 therefore does not claim a static universe is established; it asks whether a specific alternative

survives explicit tests that simpler static models fail. Failed mechanisms, null results, and adverse predictions are retained as results rather than edited away.

1 The proposition and the rule of engagement

The thesis tested here is deliberately stronger than a generic Hubble-diagram curve fit:

> **Thesis.** A non-expanding Euclidean geometry with two photon–vacuum interaction channels can reproduce the observed achromatic redshift, the approximately $(1 + z)$ temporal broadening of supernova signals, and the DES-SN5YR luminosity-distance relation without a dark-energy acceleration term. The same model must then survive independent predictions for chromatic transfer, angular distance, surface brightness, image sharpness, and the energy carried out of the coherent beam.

This paper adopts an explicit anti-confirmation-bias rule. A mechanism that fails a registered test is retired rather than rhetorically reinterpreted. An underpowered measurement is called inconclusive rather than supportive. An adverse prediction is placed in the same table as a favorable one. That rule has already changed the model: a proposed "Wien/filter flattening" explanation for apparent time dilation was tested and failed by approximately three orders of magnitude in stretch, while the preregistered DES secondary-maximum clock test returned an uninformative uncertainty interval rather than the forecast decisive measurement.

"Static" therefore means static large-scale spatial geometry, not a trivial model in which photons lose energy while all other observables remain unchanged. The forward interaction is inelastic and leaves a local transient state behind the advancing light front; the global constitutive law is stationary, while the illuminated corridor is not. The second channel removes a small wavelength-dependent fraction of photons from the image-forming beam. Keeping these jobs separate is the essential difference from classical tired-light proposals.

2 Physical motivation: nonlinear vacuum propagation without overclaiming

Quantum electrodynamics is nonlinear in sufficiently strong fields. The Euler–Heisenberg effective action predicts field-dependent electromagnetic interactions, and SLAC E-144 observed nonlinear Compton scattering and multiphoton pair-production processes using an intense laser colliding with a high-energy electron beam (Burke et al., 1997; Bamber et al., 1999). Those experiments do *not* establish the weak cosmological interaction proposed here, nor do they provide its rate. Their relevance is narrower: nonlinear electromagnetic interaction with the quantum vacuum is a physical phenomenon rather than a forbidden category.

The cosmological hypothesis is correspondingly quantitative. It requires an ultraweak forward interaction that is achromatic and direction preserving, plus a smaller non-forward channel with a specified energy dependence. Both must obey astronomical constraints on spectral-line coherence, image sharpness, polarization, and energy conservation. The present paper treats the effective transfer law as the object tested by DES; deriving its coefficients from the underlying QFD vacuum action remains a separate microscopic problem.

3 The two-channel transfer operator

3.1 Achromatic distance-redshift law

Let the forward channel produce a constant fractional photon-energy change per unit path,

$$\frac{dE}{dD} = -\frac{K}{c}E. \tag{1}$$

Then

$$1 + z = \frac{\nu_{\rm em}}{\nu_{\rm obs}} = \exp(KD/c), \qquad D(z) = \frac{c}{K}\ln(1+z). \tag{2}$$

The fractional change is independent of initial frequency, so every spectral coordinate is translated by the same factor,

$$\lambda_{\rm obs} = (1+z)\lambda_{\rm em}. \tag{3}$$

For a thermal spectral shape this moves the Wien peak to lower apparent temperature, $T_{\rm eff} = T/(1+z)$. This is a coordinate mapping of the spectrum; it does not by itself establish the correct radiance normalization.

3.2 Independent photons and the wake field

The principal objection to a static interpretation of supernova time dilation is correct for a stationary *state*: two mutually incoherent photons emitted $\Delta t_{\rm em}$ apart through an unchanged dispersionless medium arrive with the same interval. The present model therefore does not use the old single-wavepacket cycle-count argument.

Instead let $\chi(\mathbf{x}, t)$ denote the weak corridor-conditioning field left by the advancing light front, with $n^2 = 1 + \chi$. At effective-field level take

$$\mathcal{L}_A = \frac{1}{2}\left[\frac{n^2(\chi)}{c^2}(\partial_t A)^2 - (\nabla A)^2\right] + \mathcal{L}_\chi[\chi; I], \tag{4}$$

which gives

$$\frac{1+\chi(\mathbf{x},t)}{c^2}\,\partial_t^2 A - \nabla^2 A = 0. \tag{5}$$

In the eikonal limit the ray Hamiltonian is $H = c|\mathbf{k}|/n$, and along a ray

$$\frac{d\ln\omega}{dt} = -\frac{\partial \ln n}{\partial t} \equiv -W(\mathbf{x}, t). \tag{6}$$

The accumulated frequency change of the leading light front is therefore

$$\ln(1+z) = \frac{1}{c}\int W\,dl. \tag{7}$$

Now consider a second, fully incoherent photon emitted a time Δt later. Linearizing the slowness difference between the corridor state sampled by the two photons gives

$$\frac{d\ln\Delta t}{dl} = \frac{W}{c}. \tag{8}$$

Equations (7) and (8) have the same kernel, so

$$\boxed{\Delta t_{\rm obs} = (1+z)\Delta t_{\rm em}}, \qquad b = 1. \tag{9}$$

No mutual optical coherence between the photons appears in this derivation; the classical corridor state χ is the memory variable. The result is an identity of the effective wave equation, not a fit parameter.

This closes the referee's independent-photon objection only at the effective-field level. The microscopic theory still owes a response that is (i) rate-saturated rather than proportional to source intensity, (ii) achromatic, (iii) approximately constant through the burst, (iv) persistent for at least the inter-photon/event timescale, (v) finite under long illumination, and (vi) transversely non-dispersive. The required index change over a ~ 60 d transient is only of order $K\Delta t \sim 10^{-11}$ for $K \sim H_0$, but smallness does not substitute for a derivation.

3.3 Non-forward chromatic depletion

The second channel is non-forward and removes photons from the coherent beam with

$$\sigma_{\rm nf}(E) \propto E^{1/2} \propto \lambda^{-1/2}. \tag{10}$$

The physically defined optical depth is

$$\tau_\lambda(z) = \tau_\infty(\lambda)\left[1-(1+z)^{-1/2}\right], \qquad \tau_\infty(\lambda) = \tau_{\infty,B}\left(\frac{440\,\mathrm{nm}}{\lambda}\right)^{1/2}. \tag{11}$$

The magnitude attenuation is

$$\Delta m_\lambda = \frac{2.5}{\ln 10}\,\tau_\lambda. \tag{12}$$

This corrects a factor-of-two notation ambiguity in earlier drafts. The Hubble-diagram fit previously quoted as $\eta = 0.297$ in a $5/\ln 10$ convention corresponds to the physical rest-B optical depth

$$\boxed{\tau_{\infty,B} = 0.594}. \tag{13}$$

At this normalization the fraction removed from the coherent beam is about 10.3% at $z = 0.5$ and 16.0% at $z = 1$ in rest B. That energy cannot simply disappear; Section 8 treats this as a required ledger rather than an assumption.

3.4 Observer-frame transfer operator

For a source spectral-time surface $S(\lambda, t)$ the working transfer operator is

$$F_{\rm obs}(\lambda_o, t_o) = A(z)\,\exp\left[-\tau\left(\frac{\lambda_o}{1+z}, z\right)\right] S\left(\frac{\lambda_o}{1+z}, \frac{t_o}{1+z}\right). \tag{14}$$

No factor in Eq. (14) knows whether a feature was produced by nickel, cobalt, iron, silicon, or the continuum. Species-dependent emission physics sets S; the propagation remap is universal. That universality is falsifiable: any well-controlled physical clock that fails to share the same temporal remap would contradict the single-operator model.

4 A self-falsified mechanism: passbands do not manufacture $b = 1$

An initially attractive alternative was that Wien displacement, finite filter slicing, and wavelength-dependent amplitude erosion might make an intrinsically undilated light curve look broader. We tested that proposition before relying on the wake.

A full transfer calculation used the Hsiao Type Ia spectral time series (Hsiao et al., 2007), measured DECam g, r, i, z throughputs, the spectral coordinate mapping, and Eq. (11). With the time remap in Eq. (14) disabled, the recovered stretch changes by at most

$$|\Delta s| \leq 5 \times 10^{-4} \tag{15}$$

for $z \leq 0.8$ in every DES band, and the recovered peak epoch moves by at most 0.05 d. An independent inverse-injection test likewise recovered an apparent time-dilation exponent $\hat{b} = 0$ even when the chromatic opacity was amplified far above the fitted value. A time-independent amplitude weight changes a light curve's color and amplitude but cannot generate a macroscopic temporal remapping.

With the wake remap enabled, the same filter convolution recovers $(1 + z)$ stretch to within 0.05% in every band. The chromatic branch therefore has a simpler and more testable role: it changes amplitudes without materially changing timing.

The same calculation predicts peak $g - r$ shifts of approximately +0.010 mag at $z = 0.3$ and +0.024 mag at $z = 0.8$. For phase-matched broad spectral windows near rest 330 and 650 nm, the model predicts

$$Q(z) \equiv \frac{[F_{330}/F_{650}]_z}{[F_{330}/F_{650}]_{z\simeq 0}} = \exp[-\Delta\tau(z)], \tag{16}$$

with

$$Q(0.25) = 0.979, \qquad Q(0.5) = 0.964, \qquad Q(1) = 0.944. \tag{17}$$

Crucially, the propagation contribution is *phase-flat*: Eq. (11) has no supernova-phase dependence. Intrinsic line blending, recombination, and photospheric evolution vary strongly with phase, so phase-flatness is a useful confound discriminator.

5 DES-SN5YR distance comparison

5.1 Released Hubble diagram

For the DES-SN5YR v1.2 released distance vector we use the official $z_{\rm HD}/z_{\rm HEL}$ convention and the full STAT+SYS covariance. With the additive magnitude zero point analytically profiled, the static law is

$$\mu_{\rm stat}(z_{\rm HD}, z_{\rm HEL}) = 5\log_{10}[(1 + z_{\rm HEL})\ln(1 + z_{\rm HD})] \tag{18}$$

$$+ \frac{2.5}{\ln 10}\,\tau_{\infty,B}\left[1 - (1 + z_{\rm HD})^{-1/2}\right] + \mathcal{M}, \tag{19}$$

where the overall dimensional scale is absorbed in $\mathcal{M}$.

The default diagnostic sample contains 1,768 objects. Flat ΛCDM, with one fitted shape parameter Ω_m, gives $\chi^2 \simeq 1640.1$; Eq. (19), with one fitted shape amplitude equivalent to $\tau_{\infty,B} = 0.594$, gives $\chi^2 \simeq 1641.1$. Thus

$$\boxed{\Delta\chi^2 \equiv \chi^2_{\rm stat} - \chi^2_{\Lambda\rm CDM} = +1.1}, \tag{20}$$

with both $\chi^2/{\rm dof} \simeq 0.93$. Retaining all 1,829 released entries changes no quoted headline number. The maximum covariance-offset-removed separation of the two best-fit curves is about 0.055 mag over the DES sample.

This result is SN-only. It does not imply that the static model has passed BAO, CMB acoustic-scale, lensing, or growth constraints. It establishes a narrower point: the released DES-SN5YR Hubble diagram itself does not strongly distinguish the two one-parameter distance laws.

5.2 BBC sensitivity belongs in the methods layer

A companion methods analysis reconstructs a diagnostic vector before the event-level BBC magnitude correction and finds that the same pairwise statistic changes from +1.1 to +12.8, a total processing leverage of $-11.7\ \chi^2$ units. That number is *not* evidence for the static model: the uncorrected vector contains known selection bias and the released covariance belongs to the corrected product. Its role here is only to motivate a matched forward/BBC simulation under the alternative propagation law. Details, robustness probes, and continuous-integration verification are maintained separately under DOI 10.5281/zenodo.22022089.

6 Time-domain universality: what is measured and what T4 added

6.1 Existing observations

White et al. measured 1,504 DES Type Ia supernovae and found (White et al., 2024)

$$\Delta t_{\rm obs} = \Delta t_{\rm em}(1+z)^b, \qquad b = 1.003 \pm 0.005\,({\rm stat}) \pm 0.010\,({\rm sys}). \tag{21}$$

This excludes any static model that predicts no temporal remapping. Equation (9) is designed to meet that observation, so the DES result is a consistency requirement, not evidence favoring the wake over FLRW.

A second, physically different clock already exists in the literature. Blondin et al. used multi-epoch spectra of 13 high-redshift SNe Ia and found that the spectral aging rate is consistent with the expected $1/(1+z)$ scaling (Blondin et al., 2008). Again, this rejects no-dilation models but does not discriminate FLRW from a universal non-metric remap.

6.2 T4: preregistered iron-group secondary-maximum clock

We therefore preregistered an in-house test using the Type Ia i/z secondary maximum, an iron-group recombination/opacity clock rather than a single atomic line. Recombination of iron-peak material and blended Fe II emission near 7500 Å are known contributors to the I-band secondary maximum (Jack et al., 2015). The transfer model predicts that this physical feature must share the same universal time remap as the broadband envelope.

Before inspecting a usable DES result, the estimator was frozen as

$$\ln\left[\frac{\Delta t_{\rm obs}}{\Delta t_{\rm model}(z, {\rm band})}\right] = a + b_2 \ln(1+z) + \gamma(x_1 - x_{1,0}) + \epsilon, \tag{22}$$

with $b_2 = 1$ for the universal remap and $b_2 = 0$ for no dilation. The nuisance intercept absorbs template-phase normalization. The feature search used a hypothesis-neutral observer-frame window, and the preregistration specified x_1 matching, selection checks, template sensitivity, and subtype stratification where metadata permitted.

The executed DES result is

$$\boxed{b_2 = +0.52 \pm 0.78, \qquad N = 120.} \tag{23}$$

It is 0.6σ from $b_2 = 1$ and 0.7σ from $b_2 = 0$; neither kill condition fires. This is an inconclusive measurement, not evidence for an intermediate physical mixture. Frozen robustness variants gave $b_2 = 0.60$ without the x_1 term and 0.73 for $|x_1| < 1$. A labeled post-hoc inverse-variance weighting gave 1.15 ± 0.87, the same no-discrimination conclusion.

The experiment failed on power, not on the registered discriminant. The realized median uncertainty in the secondary-maximum epoch was 5.3 d rather than the forecast 2 d, intrinsic bump diversity contributed roughly an 8 d equivalent spread, and the feature leaves DES *griz* beyond about $z = 0.3$. Approximately 25 times the effective statistics would be needed for a 3σ separation of $b_2 = 0$ and 1 under the realized noise. Detection efficiency showed no significant redshift-dependent selection flag: 0.92 in the low-z bin versus 0.84 in the high-z bin, a 0.8σ difference.

The audit trail is part of the result. The first frozen implementation produced essentially no valid detections because unfiltered difference-imaging artifacts hijacked peak finding. A post-data 5σ leave-one-out epoch screen was therefore added and explicitly labeled; at that stage no valid b_2 existed, so the repair could not be tuned toward either hypothesis. Three implementation defects in GP centering, iterative outlier handling, and scale-dependent numerical jitter were likewise corrected while the quality gate still voided the regression. The final measurement is consequently retained as an honestly underpowered test rather than upgraded into support.

7 Tolman surface brightness: the present strongest threat

In Euclidean geometry with a strictly forward, transversely symmetric wake, angular size is simply

$$D_A = D = \frac{c}{K}\ln(1+z). \tag{24}$$

Before non-forward attenuation, the two redshift factors in energy and arrival rate give

$$D_{L,0} = (1+z)D, \qquad \frac{D_{L,0}}{D_A} = 1+z, \tag{25}$$

not the metric Etherington relation $(1+z)^2$. Including the non-forward optical depth,

$$I_{\mathrm{obs}} \propto (1+z)^{-2} e^{-\tau_\lambda(z)}. \tag{26}$$

It is useful to express this locally as

$$n_{\mathrm{eff}}(z,\lambda) = 2 + \frac{\tau_\lambda(z)}{\ln(1+z)}, \tag{27}$$

which is approximately 2.16–2.18 in rest I/R near the Lubin–Sandage cluster mean $z \simeq 0.86$.

Lubin and Sandage measured the Tolman signal for 34 early-type galaxies in three clusters near $z = 0.8$–0.9 and reported, in their $q_0 = 1/2$ distance frame, $n_R = 2.59 \pm 0.17$ and $n_I = 3.37 \pm 0.13$ (Lubin & Sandage, 2001a,b). The I value is often quoted as a severe exclusion of tired light. Their observed surface brightnesses and angular Petrosian radii are cosmology-independent, but placement on the local surface-brightness–linear-radius relation is not: $R = \theta D_A$ depends on the distance law. Lubin and Sandage themselves published the exponent under several q_0 choices, thereby measuring this frame sensitivity.

Before accessing a full static-frame reduction, we preregistered an analytic frame-shift diagnostic using their own q_0 grid. Regressing the published exponents against their corresponding $\log D_A$ gives slopes of approximately -3.41 per dex in R and -3.17 per dex in I. Extrapolating that measured response to Eq. (24) moves the nominal values to

$$n_R \simeq 1.45 \pm 0.17, \qquad n_I \simeq 2.30 \pm 0.13. \tag{28}$$

The apparent I-band conflict therefore becomes about 1.1σ, but the tension does not disappear: it migrates to R at about 4.3σ. Moreover, because both bands move similarly with distance reassignment, their spread is approximately frame invariant,

$$(n_I - n_R)_{\text{L\&S}} = 0.78 \pm 0.21, \tag{29}$$

whereas the present opacity law predicts about -0.02. That is a $\sim 3.7\sigma$ discrepancy before a modern systematic reanalysis.

The extrapolation is about 3.5 times wider than the distance range over which the Lubin–Sandage q_0 response was calibrated, so it cannot be called a resolution. The decisive test is preregistered: fetch the published per-galaxy Petrosian data, recompute $R = \theta D_A$ under Eq. (24), apply the two-channel K/opacity transfer consistently, forbid a cosmological lookback-time luminosity-evolution term, and refit both bands with the full error budget. The anti-cherry-pick clause requires the I improvement, the R migration, and the band spread to be reported together. If the joint (n_R, n_I) remains inconsistent with the model by more than 3σ after the full reduction, the model fails the Tolman test.

The same distance-duality difference also creates a thermal-radiance test. A source whose spectral peak has shifted to $T/(1+z)$ is not automatically a blackbody with FLRW radiance normalization; before opacity the static model is brighter by a factor $(1+z)^2$ than a true Planck surface at that shifted temperature. Expanding-photosphere measurements of Type II supernovae at moderate redshift therefore provide an independent distance/radiance discriminator.

8 Energy and image ledger

At $z = 1$ the fitted non-forward channel removes about 16% of rest-B beam photons. The theory therefore owes more than a total cross-section. It must ultimately provide a differential redistribution kernel

$$\frac{d^2\sigma}{d\Omega\, dE'} \tag{30}$$

that simultaneously closes the beam, diffuse-background, and vacuum-energy accounts.

The current working hypothesis is that non-forward events feed a prompt frequency cascade into the diffuse bath rather than leaving slightly reddened photons in a narrow optical halo. That statement is not yet a derived result. A viable kernel must pass three gates: (i) no redshift-correlated PSF wings above observational limits, (ii) no optical/near-IR extragalactic-background excess from the removed fraction, and (iii) a cascade/thermalization rate fast enough to move the scattered population out of the optical while conserving energy. The forward wake separately has zero first-order transverse gradient for a radial ray in a spherically symmetric source-conditioned corridor; any residual profile is expected to be guiding rather than diffusive. These image claims remain subordinate to a microscopic calculation of the response.

9 A purpose-built photon-counting experiment

The remaining chromatic prediction does not require a conventional classification spectrograph. Once a supernova has been discovered and its redshift measured, a purpose-built telescope can simply monitor selected spectral windows with high cadence and high relative stability.

A practical instrument would place a dichroic tree or interchangeable narrow-band filter set behind a sufficiently large collecting mirror and feed simultaneous channels to avalanche-photodiode (APD/SPAD) photon counters. For low to moderate redshift, red-enhanced silicon detectors can cover much of the required range; higher-redshift tests of a red rest-frame window require an InGaAs or comparable near-IR counting channel. The filters are chosen in the observer frame according to

$$\lambda_{\rm filt} \simeq (1+z)\lambda_{\rm rest}, \tag{31}$$

so the same rest-frame spectral windows are followed as the event evolves.

The first experiment should avoid claims about isolated "nickel lines." Type Ia spectra are broad and blended. Instead, monitor phase-matched integrated windows such as the iron-group-dominated 320–340 nm region and a red 620–680 nm reference, together with an i/z/near-IR channel containing the secondary maximum. The detector records three observables directly: count-rate ratio, feature/secondary-maximum epoch, and temporal width. Simultaneous channels, field reference stars, and an internal calibration source reduce the wide-baseline spectrophotometric systematics that limit slit spectroscopy.

The decisive chromatic signature is Eq. (17): a redshift-monotonic, approximately $\lambda^{-1/2}$, *phase-flat* suppression whose amplitude is fixed in advance by the Hubble-diagram fit. The purpose-built experiment therefore does not need exquisite centroid resolution; spectroscopy has already supplied z. It needs stable relative photon counting at the percent level over weeks to months. The same instrument can test the temporal universality with much denser cadence than DES, turning the T4 power failure into an engineering requirement rather than a conceptual ambiguity.

10 Falsification ledger

Table 1: Current status of the main tests. "Consistent" never means uniquely supportive when ΛCDM predicts the same observable.

Test	Current result	Status / kill condition
DES Hubble diagram	$\Delta\chi^2 = +1.1$ vs flat ΛCDM	Near degeneracy on released SN distances; SN-only result.
DES bulk duration	$b = 1.003 \pm 0.005 \pm 0.010$	Consistent with wake and FLRW; kills no-dilation static branch.
Blondin spectral aging	aging rate $\propto 1/(1+z)$	Independent spectral clock; consistent with universal remap and FLRW.
Filter/Wien artifact hypothesis	wake off: $\lvert\Delta s\rvert \leq 0.0005$	**Falsified as an explanation of** $b = 1$; retained as a null result.
T4 secondary maximum	$b_2 = 0.52 \pm 0.78$, $N = 120$	**Inconclusive**; contains both 0 and 1. DES *griz* underpowered.
Chromatic amplitude transfer	$Q = 0.979/0.964/0.944$ at $z = .25/.5/1$	Open discriminator; must be phase-flat and $\lambda^{-1/2}$ with no refit.
Tolman distance frame	static extrapolation: $n_I \simeq 2.30$, $n_R \simeq 1.45$	Frame shift removes nominal I crisis but creates R tension; full reduction preregistered.
Tolman band spread	0.78±0.21 observed vs −0.02 predicted	**Strongest present threat** ($\sim 3.7\sigma$); independent modern replication required.
Energy/image ledger	$\sim 16\%$ rest-B beam loss by $z = 1$	Differential kernel, EBL, halo, and cascade closure owed.
Wake microphysics	$b = 1$ derived at effective-field level	Derive $W = K$, memory/saturation, and energy storage from underlying vacuum action.

11 Discussion

The important result of this revision is not that every test favors a static interpretation. They do not. The Tolman two-band result is presently adverse; T4 is uninformative; the non-forward energy ledger is incomplete; and the wake still lacks a first-principles microscopic response coefficient. The result is that the alternative has become sufficiently specified to be wrong in several different ways.

This changes the historical tired-light comparison. A classical model with achromatic energy loss but no temporal remapping is already excluded by DES and by spectral-aging measurements. The model studied here instead contains a separate effective-field mechanism that maps mutually incoherent emission intervals by the same factor as the frequency redshift. Whether that wake exists in nature is open, but the claim is no longer "time dilation is absent." Likewise, the model does not ask chromatic erosion to create spectral redshift or time dilation: the filter/SED calculation directly falsified that possibility. The chromatic channel is left with a smaller job and a cleaner observational signature.

The paper also illustrates why null results are valuable. The failed passband-artifact explanation reduced the available theoretical freedom. The inconclusive T4 run quantified the exact cadence, diversity, and redshift leverage required for a useful species-clock experiment. The Tolman preregistration converts a historically model-dependent argument into a two-band kill condition before the full static-frame reduction is performed. These are scientifically useful outcomes even when they do not produce a preference for the proposed model.

The purpose-built photon-counting experiment follows the same logic. Rather than infer a weak chromatic transfer through continuum-warped spectra, monitor preselected rest-frame windows simultaneously and ask whether one fixed $\lambda^{-1/2}$ law explains their relative count rates while leaving timing unchanged except for the universal remap. A null at sufficient precision would kill the non-forward channel. A positive result would still not establish the entire static cosmology, but it would identify a propagation signature not contained in baseline FLRW.

12 Conclusion

A two-channel propagation model in static Euclidean geometry can now be stated in a form that is not dead on arrival against the canonical supernova objections. The forward effective wave equation gives the same integral for achromatic redshift and independent-photon temporal remapping, so $b = 1$ is an identity of the wake model rather than an optical-coherence assumption. A separate $E^{1/2}$ non-forward channel supplies the Hubble-diagram curvature through a physically defined rest-B optical depth $\tau_{\infty,B} = 0.594$ and predicts a weak, fixed chromatic transfer.

The released DES-SN5YR Hubble diagram remains nearly indifferent between this one-parameter static law and one-parameter flat ΛCDM, $\Delta\chi^2 = +1.1$. The full filter/SED calculation has independently killed the idea that amplitude erosion can fake the observed $(1 + z)$ duration: with the wake disabled, the apparent stretch remains zero to approximately 5×10^{-4}. A preregistered species-clock test on DES raw i/z photometry returned $b_2 = 0.52 \pm 0.78$ and is correctly classified as inconclusive rather than supportive.

The model's most serious current empirical challenge is no longer the headline 9σ Tolman number quoted in an FLRW radius frame. Re-expressing the published Lubin–Sandage distance sensitivity in the static frame makes the I band roughly consistent but moves the discrepancy to R, while the $R - I$ exponent spread remains about 3.7σ and cannot be removed by changing the distance law alone. The full static-frame Tolman reduction is therefore a genuine kill test, not a rhetorical appendix.

The next experimental step can also be made unusually direct: a multi-channel photon-counting telescope aimed at newly discovered supernovae can monitor fixed rest-frame spectral windows and secondary-maximum timing without continuum warping or wide-baseline slit calibration. The theory predicts what should move, what should not, and by how much. That is the standard by which the alternative should now be judged.

Data and code availability

The DES-SN5YR public light curves and v1.2 distance products are the observational basis of this work. The canonical static-model code repository, proof-ledger documents, transfer-integral producers, Tolman preregistration, and T4 execution record are maintained at github.com/tracyphasespace/Static-Universe-DES-SN5YR. This Draft 6 revision is archived at DOI 10.5281/zenodo.22031121 and supersedes Draft 5 (DOI 10.5281/zenodo.22025329) while preserving that earlier record as provenance. The independent BBC/model-discrimination methods package is archived at DOI 10.5281/zenodo.22022089 and maintained separately so its methodological result does not depend on the cosmological interpretation.

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References

Bamber, C., Boege, S. J., Koffas, T., et al. 1999, Phys. Rev. D, 60, 092004.

Blondin, S., Davis, T. M., Krisciunas, K., et al. 2008, ApJ, 682, 724, doi:10.1086/589568.

Burke, D. L., Field, R. C., Horton-Smith, G., et al. 1997, Phys. Rev. Lett., 79, 1626.

Camilleri, R., Davis, T. M., Vincenzi, M., et al. 2024, MNRAS, 533, 2615.

Hsiao, E. Y., Conley, A., Howell, D. A., et al. 2007, ApJ, 663, 1187.

Jack, D., Baron, E., & Hauschildt, P. H. 2015, MNRAS, 449, 3581, doi:10.1093/mnras/stv474.

Kessler, R., & Scolnic, D. 2017, ApJ, 836, 56.

Lubin, L. M., & Sandage, A. 2001a, AJ, 122, 1071, doi:10.1086/322133.

Lubin, L. M., & Sandage, A. 2001b, AJ, 122, 1084, doi:10.1086/322134.

Sánchez, B. O., Brout, D., Vincenzi, M., et al. 2024, ApJ, 975, 5.

Vincenzi, M., Brout, D., Armstrong, P., et al. 2024, ApJ, 975, 86.

White, R. M. T., Davis, T. M., Lewis, G. F., et al. 2024, MNRAS, 533, 3365.

Zwicky, F. 1929, Proc. Natl. Acad. Sci. USA, 15, 773.